🏥 Healthcare Operations Analytics SQL Project

Business Problem

A medium-sized hospital network is experiencing increasing operational strain due to rising patient volumes and growing complexity in care delivery.

Over the past year, leadership has observed inconsistencies in how patients move through the hospital system, from arrival to final disposition. These inconsistencies are beginning to impact both operational efficiency and patient experience, but the exact causes are unclear.

Hospital operations involve multiple moving parts:

- Varying patient inflow across time
- Department-level capacity differences
- Admission decision variability
- Trade-offs between speed and quality of care

While data is being captured across patient encounters, it has not yet been systematically analyzed to support decision-making.

Hospital leadership believes there are **hidden inefficiencies and patterns within patient flow** that are not immediately visible.

The leadership team wants the analytics team to explore the data, uncover **actionable insights and provide recommendations that directly impact hospital operations**.

Your Task

You are part of the analytics team tasked with analyzing **15 months of patient encounter data (Apr 2023 – Oct 2024)**.

Your goal is to **interrogate the data, surface meaningful insights, and translate them into actionable, data-backed recommendations** that can help hospital leadership:

- Improve operational efficiency
- Better manage patient flow
- Make more informed admission and resource decisions
- Enhance overall patient experience

Primary Business Question:

How can the hospital leverage patient flow data to identify operational inefficiencies, improve patient experience, and optimize resource allocation across departments?